Economic Resilience Under Disruption: A DIIM-Based Framework for Key Sector Identification in Singapore

Yeoh Yu Yong

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES

A DISSERTATION SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN MATHEMATICAL SCIENCES

2026

Contents

Abstract iii

Acknowledgement iv

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 The Dynamic Inoperability Input-Output Model . . . 1
- 1.3 The Key Sector Identification Problem . . . 1
- 1.4 Objectives . . . 2
- 1.5 Report Organisation . . . 2

2 Methodology 3
- 2.1 The Dynamic Inoperability Input-Output Model . . . 3
  - 2.1.1 Inoperability . . . 3
  - 2.1.2 From Leontief to IIM . . . 3
  - 2.1.3 The DIIM . . . 4
  - 2.1.4 Initial Inoperability . . . 5
  - 2.1.5 Elasticity Coefficient . . . 5
  - 2.1.6 Economic Loss . . . 5
- 2.2 Key Sector Identification Methods . . . 5
  - 2.2.1 DIIM-Based Identification (Benchmark) . . . 5
  - 2.2.2 Total Output (xi) . . . 6
  - 2.2.3 PCA $\times$ xi . . . 6
  - 2.2.4 PageRank $\times$ xi . . . 6
  - 2.2.5 Backward Linkage $\times$ xi (BL $\times$ xi) . . . 6
  - 2.2.6 Forward Linkage $\times$ xi (FL $\times$ xi) . . . 6
- 2.3 Performance Metric . . . 7

3 Case Study 1: COVID-19 Pandemic 8
- 3.1 Data and Model Parameters . . . 8
  - 3.1.1 Input-Output Table . . . 8
  - 3.1.2 Initial Inoperability . . . 8
  - 3.1.3 Model Parameters . . . 8
- 3.2 Results . . . 8
  - 3.2.1 Inoperability Evolution . . . 8
  - 3.2.2 Economic Loss Evolution . . . 9
  - 3.2.3 Identifying Vulnerable Sectors . . . 9

4 Case Study 2: Manpower Disruption 11
- 4.1 Background: Singapore's Foreign Worker Dependence . . . 11
- 4.2 Data and Model Parameters . . . 11
  - 4.2.1 Input-Output Table . . . 11

4.2.2 Initial Inoperability . . . 11
4.2.3 Model Parameters . . . 12
4.3 Results . . . 12
4.3.1 Inoperability Evolution . . . 12
4.3.2 Economic Loss Evolution . . . 12
4.4 Comparison with COVID-19 . . . 13

5 Key Sector Identification Methods **14**
5.1 PCA Applied to the IO Matrix . . . 14
5.2 Cheap Method Rankings . . . 15
5.3 Comparison at Original Disruption Data . . . 15
5.4 Effect of the Number of Key Sectors . . . 16

6 Decision Rules: When Can Cheap Methods Substitute for DIIM? **18**
6.1 Framework . . . 18
6.1.1 Motivation . . . 18
6.1.2 Performance Metric . . . 18
6.2 Monte Carlo Simulation . . . 18
6.3 COVID-19 Results . . . 19
6.4 Manpower Results . . . 19
6.5 Decision Rules via Logistic Regression . . . 20
6.6 Practical Recommendations . . . 22

7 Discussion **23**
7.1 Cross-Scenario Comparison . . . 23
7.2 Why Economic Size Dominates Pure Structure . . . 23
7.3 The Role of q_0 Concentration . . . 24
7.4 National Security Implications . . . 24
7.5 Limitations . . . 24
7.6 Future Work . . . 25

8 Conclusion **26**

Appendix A: COVID-19 Scenario Data **28**

Appendix B: Manpower Scenario Data **30**

Appendix C: Code **31**

Abstract

This project develops a quantitative framework for assessing economic resilience under disruption using the Dynamic Inoperability Input-Output Model (DIIM) [1, 2]. The DIIM captures how sectoral disruptions propagate through input-output linkages and how the economy recovers over time. Two disruption scenarios relevant to Singapore are studied: (1) the COVID-19 pandemic, using the 2019 input-output (IO) table aggregated to 15 sectors, and (2) a manpower disruption scenario motivated by potential grey zone conflicts that could trigger large-scale foreign worker departures, using the 2022 IO table with 107 sectors.

A central challenge in disruption management is identifying which sectors should receive priority intervention. The DIIM can identify these "key sectors" by simulating the full disruption and ranking sectors by economic loss, but this approach is computationally expensive and requires crisis-time data (initial inoperability q_0, demand perturbation c^*) that may not be available during the early stages of an event. This motivates the search for cheaper structural alternatives that use only the IO table.

This study evaluates five cheap methods—Total Output, PCA $\times$ xi, PageRank $\times$ xi, BL $\times$ xi, and FL $\times$ xi—against the DIIM benchmark through Monte Carlo simulation (2 000 random q_0 vectors per scenario). Performance is measured by the ratio of economic loss reduction achieved by each cheap method relative to DIIM. For the 15-sector COVID scenario, the best structural methods (Total Output, PageRank $\times$ xi, BL $\times$ xi, and FL $\times$ xi) match DIIM in 89.8% of cases, and PCA $\times$ xi matches in 86.6%. For the 107-sector manpower scenario, FL $\times$ xi and Total Output match in 55.4% of cases, while PCA $\times$ xi drops to 38.8%. Logistic regression is used to derive decision rules that predict, from observable q_0 features, when each cheap method will perform adequately.

Acknowledgement

I would like to express my sincere gratitude to my supervisor(s) for their guidance, encouragement, and constructive feedback throughout this project. Their expertise and patience were invaluable in shaping the direction of the work and in helping me develop my research and writing skills.

I am also grateful to the members of my committee and the faculty in the department for their insightful comments and support. I would like to acknowledge the technical and administrative staff whose assistance made it possible to carry out the practical aspects of this work.

I would like to thank my colleagues and friends for the discussions, collaboration, and encouragement that made the process both productive and enjoyable. Their willingness to share ideas and offer help at key moments is deeply appreciated.

Finally, I am profoundly thankful to my family for their unwavering support, understanding, and motivation. Their confidence in me has been a constant source of strength throughout my studies.

Chapter 1

Introduction

1.1 Background

Singapore is a small, open, city-state economy whose prosperity depends on deep integration with global supply chains and a large foreign workforce. Approximately one-third of Singapore's labour force consists of non-resident workers, spanning construction, manufacturing, marine shipyard, and services [3]. This dependence, while a source of economic dynamism, also constitutes a structural vulnerability: any event that rapidly reduces the available workforce—whether a pandemic, a geopolitical crisis, or a grey zone conflict—can propagate through interconnected sectors and inflict cascading economic losses far beyond the initially affected industries.

The COVID-19 pandemic demonstrated this vividly. When Singapore imposed the "circuit breaker" lockdown from 7 April to 1 June 2020 [4, 5], the disruption was not confined to directly affected sectors. Construction shutdowns deprived building-material manufacturers of demand, transport restrictions disrupted logistics, and hospitality closures cascaded into food supply chains. Understanding these cascading effects requires a framework that captures how sectors depend on each other and how disruptions propagate over time.

Beyond pandemics, Singapore faces national security scenarios that could produce similar or even more severe manpower disruptions. Grey zone conflicts—coercive actions below the threshold of conventional warfare, such as economic coercion, cyber attacks, or hybrid threats—could trigger rapid departures of foreign workers, disrupting sectors that rely heavily on non-resident labour [6]. Unlike COVID-19, where the external demand shock (c^*) was significant, a manpower disruption scenario is characterised by high initial inoperability (q_0) with $c^* = 0$, representing a purely structural disruption that propagates through the production network.

1.2 The Dynamic Inoperability Input-Output Model

The Inoperability Input-Output Model (IIM) and its dynamic extension (DIIM) provide a rigorous framework for modelling such disruptions [1]. Built on the Leontief input-output model, the DIIM measures "inoperability"—the normalised production loss of each sector on a scale from 0 (normal) to 1 (complete shutdown)—and simulates how initial shocks propagate through input-output linkages and how the economy recovers over time. The Shanghai COVID-19 study demonstrated the utility of this approach for quantifying sectoral vulnerability and evaluating risk management strategies [2].

1.3 The Key Sector Identification Problem

A critical application of the DIIM is identifying "key sectors"—those where targeted intervention (e.g., reducing inoperability by allocating recovery resources) yields the greatest reduction in total economic

loss. The DIIM identifies these sectors by running the full dynamic simulation and ranking sectors by cumulative economic loss. However, this approach has two limitations:

1. **Computational cost**: The DIIM requires running the complete simulation for every candidate intervention strategy.
2. **Data requirements**: The DIIM needs crisis-time data—initial inoperability q_0 and the demand perturbation c^*—which may not be available in the early stages of a disruption or during pre-event planning.

This motivates the search for *cheap structural methods* that can identify key sectors using only the input-output table, which is available in peacetime. Principal Component Analysis (PCA) applied to the integrated IO matrix $H = A(I-A)^{-1}$ is one such method, previously shown to produce sector rankings comparable to DIIM for specific scenarios. However, whether PCA's performance generalises across different disruption profiles (i.e., different q_0 vectors) is an open question.

1.4 Objectives

This study addresses the following objectives:

1. Apply the DIIM framework to two disruption scenarios relevant to Singapore: the COVID-19 pandemic (15 sectors, 2019 IO table) and a manpower disruption (107 sectors, 2022 IO table).
2. Evaluate five cheap structural methods for key sector identification—Total Output, PCA $\times$ xi, PageRank $\times$ xi, BL $\times$ xi, and FL $\times$ xi—against the DIIM benchmark.
3. Determine the boundary conditions (in terms of the initial inoperability vector q_0) under which each cheap method can reliably substitute for the full DIIM simulation.
4. Derive practical decision rules that a policymaker can use to decide, given observable features of q_0, whether to use a cheap method or invest in running the full DIIM.

1.5 Report Organisation

Chapter 2 presents the methodology, including the derivation of the DIIM and the formulation of the cheap structural methods. Chapter 3 applies the DIIM to the COVID-19 scenario, presenting results and risk management analysis. Chapter 4 extends the analysis to the manpower disruption scenario. Chapter 5 compares the key sector identification methods across both scenarios. Chapter 6 develops the decision rules framework through Monte Carlo simulation and logistic regression. Chapter 7 discusses the implications, and Chapter 8 concludes.

Chapter 2

Methodology

This chapter presents the analytical framework. Section 2.1 derives the DIIM from the Leontief input-output model. Section 2.2 introduces the cheap structural methods for key sector identification. Section 2.3 defines the performance metric used to compare methods.

2.1 The Dynamic Inoperability Input-Output Model

2.1.1 Inoperability

This paper uses the DIIM model to assess the impacts of economic disruptions on the inoperability and economic losses of Singapore's sectors. The DIIM model is derived from the Leontief input-output model and is used to investigate the higher-order transmission effect of input-output linkages between sectors on inoperability. Using the DIIM model requires first introducing the concept of "inoperability", which is defined as follows:

$$\text{Inoperability} = \frac{\text{As Planned Production} - \text{Degraded Production}}{\text{As Planned Production}} \tag{2.1}$$

Where "As Planned Production" represents the output level of a sector under normal production, and "Degraded Production" represents the output level of a sector after being shocked in its production process. "Inoperability" takes values between 0 and 1, with higher values indicating greater damage to production caused by the shock. A value of 1 means the shocked sector has completely lost its production capacity, while a value of 0 means the sector is producing at a normal level.

2.1.2 From Leontief to IIM

Using the concept of inoperability defined above, the IIM model can be derived from the Leontief input-output model, as shown in Equation (2.2).

$$x = Ax + c \tag{2.2}$$

Where x is the total output vector; A is the technical coefficient matrix; c is the final demand vector. If we define the output levels and the final demand vector of some shocked sectors as $\tilde{x}$ and $\tilde{c}$ respectively, then we can construct the input-output relationship between sectors after the shock.

Subtracting Equation (2.2) from the shocked-economy formulation:

$$x - \tilde{x} = A(x - \tilde{x}) + (c - \tilde{c}) \tag{2.3}$$

Defining $\hat{x}$ as a diagonalized matrix of the output vector and left multiplying $\hat{x}^{-1}$ on both sides of Equation (2.3):

$$\hat{x}^{-1}(x - \tilde{x}) = \hat{x}^{-1}A(x - \tilde{x}) + \hat{x}^{-1}(c - \tilde{c}) \tag{2.4}$$

Let $q = \hat{x}^{-1}(x - \tilde{x})$, $\quad A^* = \hat{x}^{-1} A \hat{x}$, $\quad c^* = \hat{x}^{-1}(c - \tilde{c})$; then Equation (2.5) can be derived, which is the IIM model.

$$q = A^* q + c^* \tag{2.5}$$

The details of each matrix in the IIM model are shown in Equations (2.6), (2.7), and (2.8):

$$q = \hat{x}^{-1}(x - \tilde{x}) = \begin{bmatrix} \frac{1}{x_1} & 0 & \cdots & \cdots & 0 \\ 0 & \ddots & & & \vdots \\ \vdots & & \frac{1}{x_i} & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} x_1 - \tilde{x}_1 \\ \vdots \\ x_i - \tilde{x}_i \\ \vdots \\ x_n - \tilde{x}_n \end{bmatrix} \tag{2.6}$$

$$A^* = \hat{x}^{-1} A \hat{x} = \begin{bmatrix} a_{11}\frac{x_1}{x_1} & \cdots & a_{1j}\frac{x_j}{x_1} & \cdots & a_{1n}\frac{x_n}{x_1} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1}\frac{x_1}{x_i} & \cdots & a_{ij}\frac{x_j}{x_i} & \cdots & a_{in}\frac{x_n}{x_i} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1}\frac{x_1}{x_n} & \cdots & a_{nj}\frac{x_j}{x_n} & \cdots & a_{nn}\frac{x_n}{x_n} \end{bmatrix} \tag{2.7}$$

$$\mathbf{c}^* = \hat{\mathbf{x}}^{-1}(\mathbf{c} - \tilde{\mathbf{c}}) = \begin{bmatrix} \frac{1}{x_1} & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \frac{1}{x_i} & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} c_1 - \tilde{c}_1 \\ \vdots \\ c_i - \tilde{c}_i \\ \vdots \\ c_n - \tilde{c}_n \end{bmatrix} \tag{2.8}$$

2.1.3 The DIIM

The IIM model can be extended to the DIIM model by introducing a time variable and an elasticity coefficient matrix describing the recovery capacity of sectors. The discrete form of the DIIM model is:

$$q(t+1) = q(t) + K\left[A^* q(t) + c^*(t) - q(t)\right] \tag{2.9}$$

Where K is the elasticity coefficient matrix describing the recovery capacity of sectors after being shocked in production. Assuming that the recovery capacity of a sector depends only on its own production, and is unrelated to the production linkages with other sectors, thus K is a diagonal matrix with diagonal elements greater than 0. The larger the diagonal elements, the stronger the recovery capacity of the corresponding sector in response to shocks. t is the discrete time variable, and $q(t)$ represents the sectoral inoperability vector at time t. Approximating Equation (2.9) to differential form:

$$\dot{q}(t) = K\left[A^* q(t) + c^*(t) - q(t)\right] \tag{2.10}$$

Solving Equation (2.10) we can get the equation describing the evolution of sectoral inoperability over time:

$$q(t) = e^{-K(I-A^*)t} q(0) + \int_0^t K e^{-K(I-A^*)(t-z)} c^*(z)\, dz \tag{2.11}$$

Assuming that the demand shock c^* remains unchanged, and $c^* = 0$, then Equation (2.11) can be simplified to:

$$q(t) = e^{-K(I-A^*)t} q(0) \tag{2.12}$$

Where $q(0)$ represents the initial inoperability vector of sectors after being shocked. As time goes by, the inoperability changes at a rate of $e^{-K(I-A^*)t}$.

2.1.4 Initial Inoperability

The initial inoperability $q(0)$ is calculated as:

$$\text{Sector Initial Inoperability} = \frac{\text{Unavailable Workforce}}{\text{Size of Workforce}} \times \frac{\text{LAPI}}{\text{Sector Output}} \tag{2.13}$$

where the left factor is the proportion of unavailable workers in each sector and the right factor captures how dependent each sector is on labour as an input (Local Area Personal Income).

2.1.5 Elasticity Coefficient

According to Equation (2.12), the inoperability of sector i at time t is:

$$q_i(t) = q_i(0) e^{-k_i(1-a_{ii}^*)t} \tag{2.14}$$

From Equation (2.14), we can derive the calculation formula for k_i:

$$k_i = \frac{\ln\left[q_i(0)/q_i(T)\right]}{T_i\left(1-a_{ii}^*\right)} \tag{2.15}$$

Where $q_i(0)$ is the initial inoperability of sector i; $q_i(T)$ is the inoperability of sector i at time T; a_{ii}^* is the element on the diagonal of matrix A^*, representing the self-dependence of sector i.

2.1.6 Economic Loss

Under the analytical framework of the DIIM model, the cumulative economic loss for sector i is:

$$EL_i = x_i \int_{t=0}^{t=T} q_i(t)\, dt \tag{2.16}$$

Where x_i is the planned output of sector i under normal production.

2.2 Key Sector Identification Methods

A key sector is defined as one where targeted intervention—modelled as a 10% reduction in initial inoperability q_0—produces the largest reduction in total economic loss. Below we describe the expensive benchmark and five cheap structural alternatives.

2.2.1 DIIM-Based Identification (Benchmark)

The DIIM identifies key sectors by running the full simulation without intervention, then ranking sectors by their cumulative economic loss EL_i from Equation (2.16). The top-k sectors are selected for intervention. This is the "gold standard" but requires knowledge of q_0 and c^*.

2.2.2 Total Output (xi)

This is the simplest baseline method. Sectors are ranked simply by their total planned economic output x_i. This method ignores the network structure entirely and assumes that intervening in the largest economic sectors will yield the greatest reduction in loss.

2.2.3 PCA $\times$ xi

Instead of using A directly, this approach constructs the integrated input-output coefficient matrix:

$$H = A(I - A)^{-1} = A \cdot L \tag{2.17}$$

where $L = (I - A)^{-1}$ is the Leontief inverse. The matrix H captures both direct and indirect effects through the production network. PCA is applied to H, and a structural distance is computed for each sector based on its Euclidean distance from the origin in the space of the principal components:

$$d_i^{\text{PCA}} = \sqrt{\sum_{k=1}^{n_pcs} v_{ik}^2} \tag{2.18}$$

where v_{ik} is the loading of sector i on principal component k. To account for economic size, this structural distance is weighted by the sector's total output x_i:

$$\text{Score}_i^{\text{PCA}} = d_i^{\text{PCA}} \times x_i \tag{2.19}$$

Sectors with larger $\text{Score}_i^{\text{PCA}}$ are ranked higher.

2.2.4 PageRank $\times$ xi

The interdependency matrix A^* is treated as the weighted adjacency matrix of a directed graph. The PageRank algorithm is applied to compute a structural centrality score PR_i for each sector, which measures importance based on the structure of incoming links with a damping factor. This centrality score is then weighted by the sector's total output:

$$\text{Score}_i^{\text{PR}} = PR_i \times x_i \tag{2.20}$$

2.2.5 Backward Linkage $\times$ xi (BL $\times$ xi)

Backward linkages measure the extent to which a sector draws on other sectors' outputs as inputs. They are calculated as the column sums of the Leontief inverse:

$$BL_j = \sum_{i=1}^{n} l_{ij} \tag{2.21}$$

where l_{ij} are elements of L. A high BL_j indicates that sector j generates large demand for inputs from other sectors. To identify key sectors, this structural linkage is weighted by total output:

$$\text{Score}_j^{\text{BL}} = BL_j \times x_j \tag{2.22}$$

2.2.6 Forward Linkage $\times$ xi (FL $\times$ xi)

Forward linkages measure how much a sector's output is used as inputs by other sectors. They are calculated as the row sums of the Leontief inverse:

$$FL_i = \sum_{j=1}^{n} l_{ij} \tag{2.23}$$

A high FL_i indicates that sector i is an important supplier to the rest of the economy. This structural linkage is weighted by total output:

$$\text{Score}_i^{\text{FL}} = FL_i \times x_i \tag{2.24}$$

2.3 Performance Metric

To compare a cheap method against DIIM, we define the *performance ratio*:

$$R_m = \frac{\Delta EL_m}{\Delta EL_{\text{DIIM}}} \tag{2.25}$$

where ΔEL_m is the total economic loss reduction achieved by intervening in the top-k sectors identified by cheap method m, and ΔEL_{DIIM} is the reduction achieved by the DIIM's top-k sectors. If $R_m \geq 0.95$, the cheap method is considered "close enough" to substitute for DIIM.

Chapter 3

Case Study 1: COVID-19 Pandemic

This chapter applies the DIIM framework to Singapore's COVID-19 pandemic, presenting the data, simulation results, and risk management analysis.

3.1 Data and Model Parameters

3.1.1 Input-Output Table

The IO transaction data for Singapore is obtained from the Department of Statistics, Singapore [7]. This study uses the 2019 IO table because it is the closest year before COVID-19 and represents the pre-pandemic baseline. The available IO table is aggregated into 20 broad sectors, and this study further aggregates them into 15 sectors to match the Ministry of Manpower dataset used to compute initial inoperability. The mapping is provided in Appendix A (Table 8.1).

From the aggregated IO table, the technical coefficient matrix A is computed. A cross-section of A is provided in Appendix A (Table 8.2).

3.1.2 Initial Inoperability

Initial inoperability is calculated using Equation (2.13). The workforce unavailability component is obtained from Ministry of Manpower data, and the labour dependence component is derived from the IO table. The initial inoperability values for the 15 aggregated sectors are listed in Appendix A (Table 8.3).

3.1.3 Model Parameters

Singapore's lockdown measures ("circuit breaker") were implemented from 7 April to 1 June 2020, which is 55 days [4, 5]. This study sets the shock duration to 55 days.

In late April 2022 (751 days after the start of lockdown), Singapore's DORSCON level was lowered from Orange to Yellow, and employees were allowed to return to workplaces. This study assumes a final inoperability $q_i(T_i) = 1\%$, meaning economic activity returns to 99% of pre-lockdown levels after 751 days.

3.2 Results

3.2.1 Inoperability Evolution

Figure 3.1 shows the simulated inoperability trajectories for the 15 aggregated sectors over the shock-and-recovery horizon. Inoperability rises during the shock window as disruptions propagate through input-output linkages, and it declines during recovery as sectors gradually restore production. Because sectors are interconnected, each sector experiences both direct effects from its own disruption and indirect effects transmitted from other sectors.

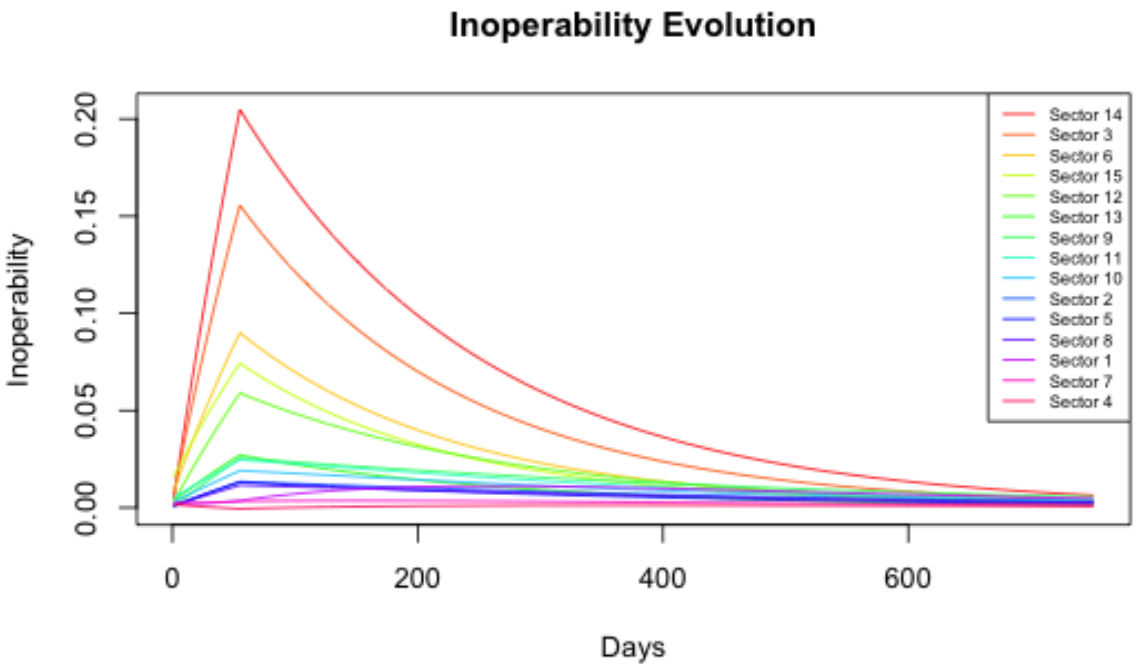

Figure 3.1: Inoperability evolution over the shock-and-recovery horizon (COVID-19, 15 sectors).

3.2.2 Economic Loss Evolution

Figure 3.2 presents the corresponding economic-loss evolution. Sectors with larger baseline outputs can accumulate large absolute losses even if their relative inoperability is not the highest, while smaller sectors may show high inoperability but smaller absolute losses.

3.2.3 Identifying Vulnerable Sectors

This section identifies vulnerable sectors using two measures: inoperability (relative production loss) and cumulative economic loss (absolute output loss). Figure 3.3 plots each sector using its rank in inoperability (y-axis) and its rank in economic loss (x-axis), so sectors that rank high in both measures are easy to identify.

To support prioritisation, we define ordinal "zones" (Top 5, Top 7, Top 10) as the intersection of sectors that appear in the top-k rankings of both measures. For example, the Top 5 zone contains sectors that are simultaneously ranked in the top five for both inoperability and economic loss, highlighting sectors that are both heavily disrupted and economically important.

Differences between the two rankings are expected. For example, a sector such as Arts, Entertainment & Recreation may rank high in inoperability but lower in economic loss if its baseline output is smaller than that of other sectors. Overall, Figure 3.3 provides a practical view for decision-making when policymakers want to balance restoring functionality (low inoperability) and reducing macroeconomic damage (low loss).

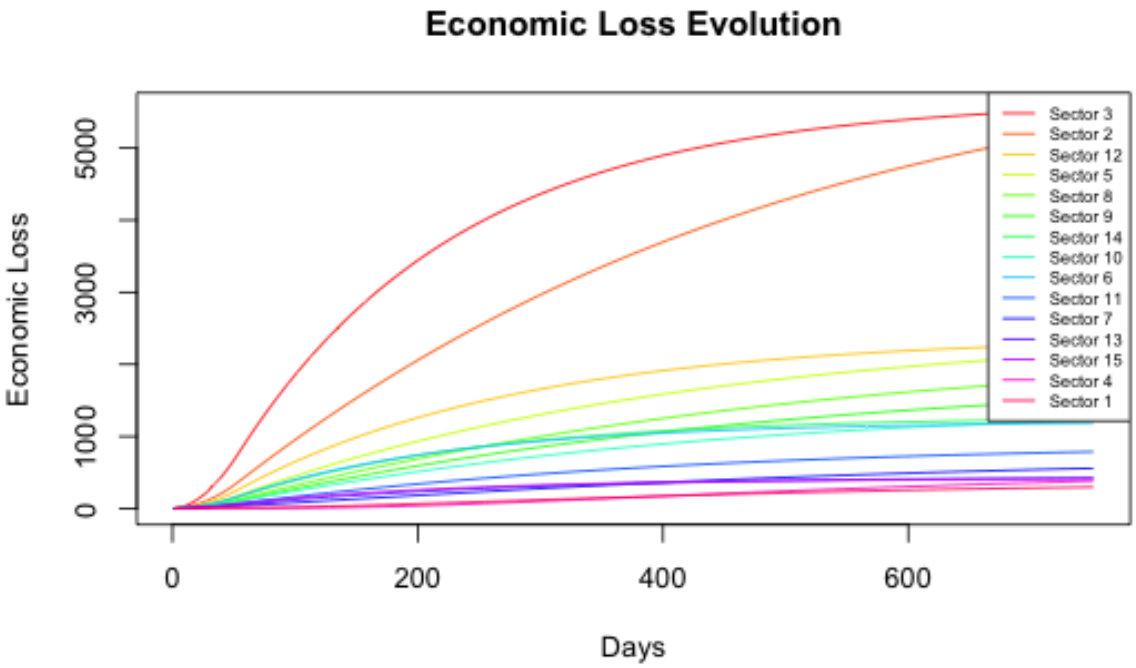

Figure 3.2: Economic loss evolution over the shock-and-recovery horizon (COVID-19, 15 sectors).

	2	3	5	12	8	9	10	14	6	11	7	13	15	4	1
14								X							
3		X													
6									X						
12				X											
15													X		
9						X									
13												X			
11										X					
2	X														
10							X								
5			X												
8					X										
1															X
7											X				
4														X	

Top 5 Zone	
3	Construction
12	Public Administration & Education

Top 7 Zone	
9	Real Estate Services

Top 10 Zone	
14	Arts, Entertainment & Recreation
6	Accommodation & Food Services
11	Administrative & Support Services
2	Manufacturing
10	Professional Services

Figure 3.3: Joint impact matrix: inoperability rank vs. economic loss rank (COVID-19).

Chapter 4

Case Study 2: Manpower Disruption

This chapter extends the DIIM framework to a manpower disruption scenario motivated by Singapore's dependence on foreign labour and the potential for grey zone conflicts to trigger workforce departures.

4.1 Background: Singapore's Foreign Worker Dependence

Singapore's economy relies on approximately 1.4 million foreign workers, who constitute roughly one-third of the total labour force [3]. These workers are concentrated in sectors such as construction, manufacturing, marine shipyard, process industries, and domestic services. While this reliance on foreign labour has been a key enabler of economic growth, it creates a structural vulnerability to events that could trigger rapid workforce departures.

Grey zone conflicts—coercive actions that fall below the threshold of conventional warfare—represent a plausible threat scenario [6]. Such conflicts could include economic sanctions, cyber attacks on critical infrastructure, disinformation campaigns, or military posturing in the region. Any of these could erode confidence among foreign workers and their home governments, potentially triggering a significant and rapid outflow of non-resident workers.

Unlike the COVID-19 scenario, where disruption arose from both workforce unavailability ($q_0 > 0$) and reduced demand ($c^* > 0$), a manpower disruption is characterised by:

- **High initial inoperability** ($q_0 > 0$): Sectors lose productive capacity as workers depart.
- **No external demand shock** ($c^* = 0$): The disruption is purely structural. Demand may persist but cannot be met due to workforce shortages.

This distinction is important because the DIIM solution simplifies to $q(t) = e^{-K(I-A^*)t}\, q(0)$ when $c^* = 0$ (Equation (2.12)), meaning the disruption dynamics are governed entirely by the structure of the economy (A^*) and the initial shock (q_0).

4.2 Data and Model Parameters

4.2.1 Input-Output Table

This scenario uses the 2022 Singapore IO table, aggregated to 15 broad sectors matching the same sectoral classification as the COVID-19 scenario. This aggregation facilitates direct comparison between the two case studies, while still capturing the heterogeneous distribution of foreign workers across sectors.

4.2.2 Initial Inoperability

Initial inoperability is computed using the same method as Equation (2.13), with workforce unavailability data reflecting the proportion of foreign workers in each sector. Sectors with high foreign worker concentrations (e.g., construction, process industries) have correspondingly higher initial inoperability.

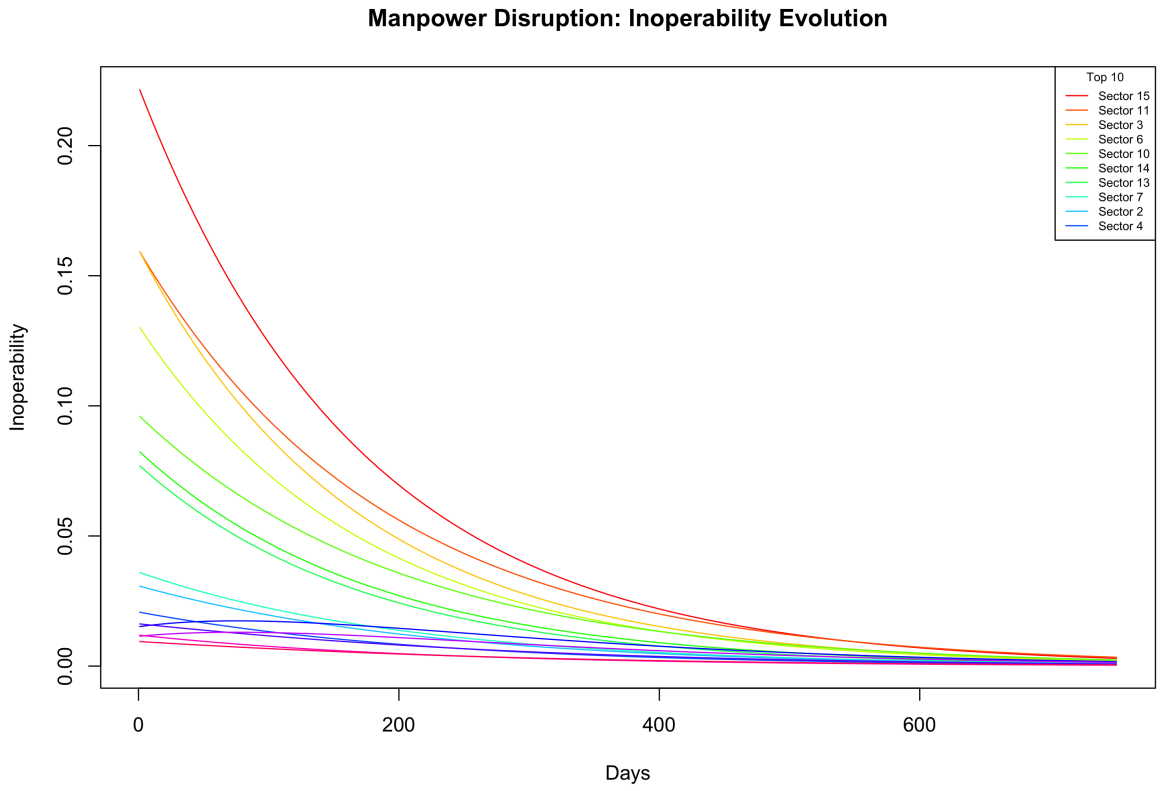

Figure 4.1: Inoperability evolution for the manpower disruption scenario (15 sectors).

4.2.3 Model Parameters

The shock duration is set to 55 days to enable comparison with the COVID-19 scenario, though a manpower disruption could have a different timeline. The total simulation horizon is 751 days with a final inoperability of 1%, representing near-full recovery through workforce replacement, retraining, or automation.

4.3 Results

4.3.1 Inoperability Evolution

Figure 4.1 shows the simulated inoperability trajectories for the 15 sectors. Compared to the COVID-19 scenario, the manpower disruption exhibits:

- Greater dispersion in initial inoperability across sectors, reflecting the uneven distribution of foreign workers.
- Purely structural cascading effects through the production network, as disruptions in labour-intensive sectors propagate to downstream industries without any external demand shock.

4.3.2 Economic Loss Evolution

Figure 4.2 presents the cumulative economic loss evolution. The 15-sector model reveals that even sectors with moderate inoperability can accumulate substantial losses if their baseline output is large (e.g., manufacturing, financial services).

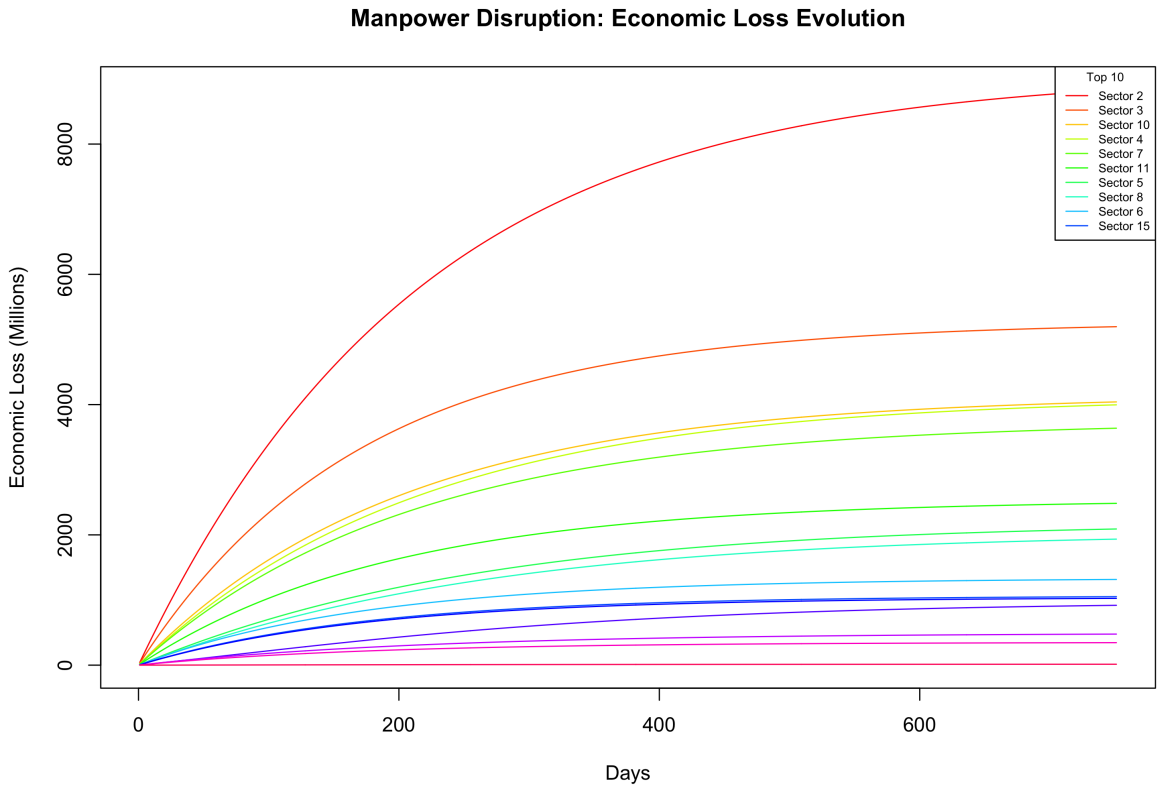

Figure 4.2: Economic loss evolution for the manpower disruption scenario (15 sectors).

4.4 Comparison with COVID-19

Table 4.1 summarises the key structural differences between the two scenarios.

Although both scenarios operate at the same 15-sector level, the manpower scenario is structurally distinct because the absence of a demand shock ($c^* = 0$) means the entire disruption propagation is driven by the production network alone. Furthermore, the 2022 IO table captures post-pandemic structural shifts in the economy. These structural differences have important implications for the effectiveness of cheap key sector identification methods, as explored in the following chapters.

Parameter	**COVID-19**	**Manpower**
Number of sectors	15	15
IO table year	2019	2022
Demand shock c^*	> 0	$= 0$
Shock type	Demand + supply	Pure supply
Lockdown duration	55 days	55 days
Total horizon	751 days	751 days

Table 4.1: Comparison of COVID-19 and manpower disruption scenario parameters.

Chapter 5

Key Sector Identification Methods

This chapter compares the key sector identification methods described in Chapter 2 using the actual disruption data from both case studies. Recall that the DIIM identifies key sectors by running the full simulation and ranking by economic loss (expensive, requires q_0 and c^*), while the cheap methods use only the IO table structure.

5.1 PCA Applied to the IO Matrix

PCA is applied to the integrated IO matrix $H = A(I-A)^{-1}$. Table 5.1 lists the PCA loadings for the COVID-19 scenario (15 sectors), sorted by Euclidean distance in the PC space, along with the final weighted score $\text{Score}_i^{\text{PCA}}$ which determines the ranking.

Sector ID	PC1	PC2	d_i^{PCA}	$\text{Score}_i^{\text{PCA}}$
2	0.489	0.548	0.734	262,791
5	0.231	0.602	0.645	110,877
4	0.164	0.198	0.257	55,084
8	0.174	0.320	0.364	49,794
3	0.773	−0.159	0.789	49,288
7	0.107	0.232	0.256	26,141
10	0.141	0.185	0.233	14,362
9	0.066	0.120	0.137	7,513
11	0.100	0.154	0.184	5,862
1	0.082	0.203	0.219	3,986
12	0.021	0.034	0.040	2,271
6	0.019	0.040	0.044	1,028
15	0.010	0.016	0.019	179
13	0.002	0.004	0.005	106
14	0.004	0.006	0.008	77

Table 5.1: PCA loadings, Euclidean distance, and weighted score $\text{Score}_i^{\text{PCA}}$ for COVID-19 sectors, sorted by weighted score.

The top-5 PCA $\times$ xi sectors for COVID-19 are sectors 2 (Manufacturing), 5 (Transportation & Storage), 4 (Wholesale & Retail Trade), 8 (Financial & Insurance Services), and 3 (Construction). While sector 3 has the largest structural distance, sector 2 becomes the highest-ranked when weighted by its massive economic size.

For the manpower scenario (15 sectors), PCA is computed on the corresponding H matrix from the 2022 aggregated IO table. The top-5 PCA $\times$ xi sectors are sectors 2, 5, 4, 7, and 8. These closely mirror

the COVID-19 PCA sectors, reflecting the similar aggregated structure, though the different IO table vintage (2022 vs 2019) and the absence of demand shock ($c^* = 0$) produce some ranking differences.

5.2 Cheap Method Rankings

Table 5.2 and Table 5.3 show the top-5 sectors identified by each method for the two scenarios.

Method	**Top-5 Sectors (COVID-19)**
Total Output	2, 4, 5, 8, 7
PCA × xi	2, 5, 4, 8, 3
PageRank × xi	2, 4, 5, 8, 7
BL × xi	2, 4, 5, 8, 7
FL × xi	2, 4, 5, 8, 7

Table 5.2: Top-5 sectors by each cheap method (COVID-19, 15 sectors).

Method	**Top-5 Sectors (Manpower)**
Total Output	2, 4, 5, 7, 8
PCA × xi	2, 5, 4, 7, 8
PageRank × xi	4, 2, 5, 7, 8
BL × xi	2, 5, 4, 7, 8
FL × xi	2, 5, 4, 7, 8

Table 5.3: Top-5 sectors by each cheap method (Manpower, 15 sectors).

Several observations emerge:

1. **High method agreement**: By incorporating total output x_i as a weight, the methods converge strongly. For COVID-19, Total Output, PageRank × xi, BL × xi, and FL × xi produce the exact same top-5 ranking (2, 4, 5, 8, 7). PCA × xi differs only slightly by including sector 3 instead of 7.

2. **Economic size dominates**: The fact that structurally distinct measures (centrality, backward linkage, and forward linkage) yield nearly identical rankings once weighted by x_i indicates that a sector's baseline economic size is the dominant driver of its overall impact in these aggregated models.

3. **Cross-scenario consistency**: Despite using different IO tables (2019 vs 2022) and different shock structures ($c^* > 0$ vs $c^* = 0$), both scenarios produce very similar cheap method rankings. The manpower scenario rankings (2, 4/5, 5/4, 7, 8) closely mirror the COVID-19 rankings (2, 4, 5, 8, 7), confirming that economic size is the overwhelmingly dominant factor in these 15-sector models.

5.3 Comparison at Original Disruption Data

Using the actual q_0 data from each scenario, we compare each cheap method's economic loss reduction against the DIIM benchmark when intervening in the top-5 sectors.

For the COVID-19 scenario, the identical rankings of Total Output, PageRank × xi, BL × xi, and FL × xi achieve a loss reduction very close to the DIIM benchmark, while PCA × xi performs slightly better. For the manpower scenario, the gap between the cheap methods and DIIM widens significantly, with even the best cheap method falling substantially short of DIIM's reduction.

This observation motivates the Monte Carlo analysis in Chapter 6: the performance of cheap methods at a single q_0 point is insufficient to draw general conclusions. We need to understand how performance varies across the space of possible q_0 vectors.

5.4 Effect of the Number of Key Sectors

Figure 5.1 shows how the economic loss reduction varies across all methods as the number of key sectors k increases, for the COVID-19 scenario. Across all methods, increasing k improves the loss reduction, but the rate of improvement differs. The new baseline, Total Output, tracks very closely with the other structurally-weighted methods, confirming that weighting by economic size captures the bulk of the necessary information at this scale.

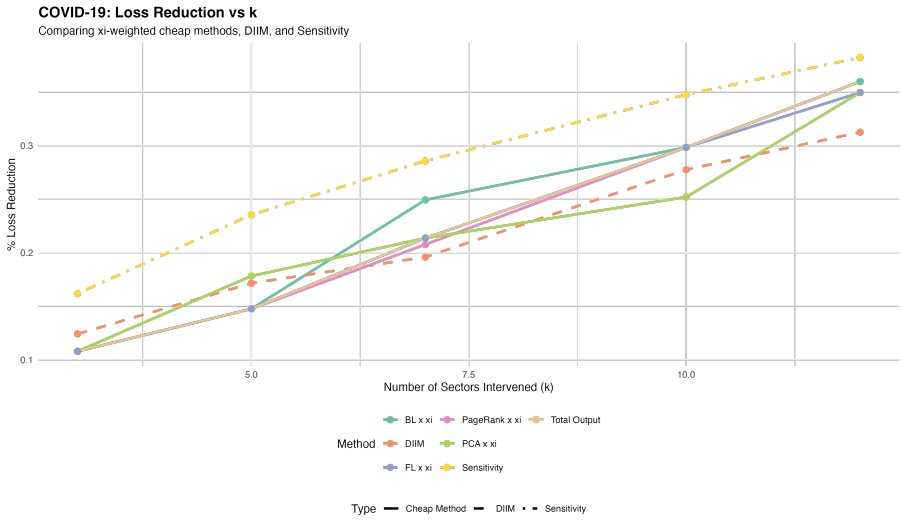

Figure 5.1: Economic loss reduction for different methods across top-k sectors (COVID-19).

Figures 5.2 and 5.3 compare the standard PCA-based and DIIM-based approaches specifically.

Figure 5.2: Economic loss reduction for PCA vs DIIM (COVID-19).

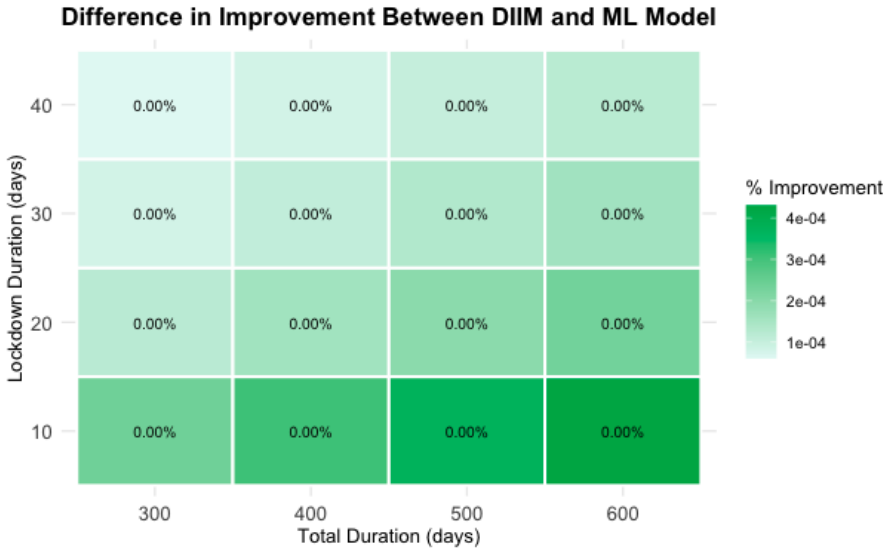

Figure 5.3: Difference in improvement: PCA vs DIIM (COVID-19).

These results demonstrate that PCA can perform well at the original COVID-19 data point, but the question remains: does this hold across a wider range of disruption profiles? This is addressed in the next chapter.

Chapter 6

Decision Rules: When Can Cheap Methods Substitute for DIIM?

The previous chapter showed that cheap methods can match DIIM at specific q_0 points but not universally. This chapter develops a systematic framework to determine *when* each cheap method can be trusted, using Monte Carlo simulation and logistic regression.

6.1 Framework

6.1.1 Motivation

All cheap methods produce a *fixed* sector ranking—they depend only on the IO table structure (A, x), not on the disruption profile (q_0). The DIIM, by contrast, adapts its ranking to the specific q_0. A cheap method performs well when its fixed ranking happens to align with where q_0 concentrates; it fails when the disruption hits sectors that the cheap method did not prioritise.

This insight suggests that the effectiveness of a cheap method is predictable from features of q_0. If we can identify these features, we can construct decision rules: "Use cheap method m if [condition on q_0]; otherwise, run the full DIIM."

6.1.2 Performance Metric

For each cheap method m and a given q_0, we compute the performance ratio from Equation (2.25). Method m is considered "close enough" if $R_m \geq 0.95$.

6.2 Monte Carlo Simulation

We draw 2 000 random q_0 vectors for each scenario (COVID-19 and Manpower, both 15 sectors). For each trial:

1. Draw $q_{0,i} \sim \text{Uniform}(10^{-6},\ 2\max(q_0^{\text{actual}}))$ independently for each sector i.
2. Run the DIIM baseline (no intervention) and compute total economic loss.
3. Run the DIIM with intervention in the DIIM's top-5 sectors (10% q_0 reduction) and compute ΔEL_{DIIM}.
4. For each cheap method m, run the DIIM with intervention in method m's top-5 sectors and compute ΔEL_m.
5. Compute $R_m = \Delta EL_m / \Delta EL_{\text{DIIM}}$ and record whether $R_m \geq 0.95$ ("close").

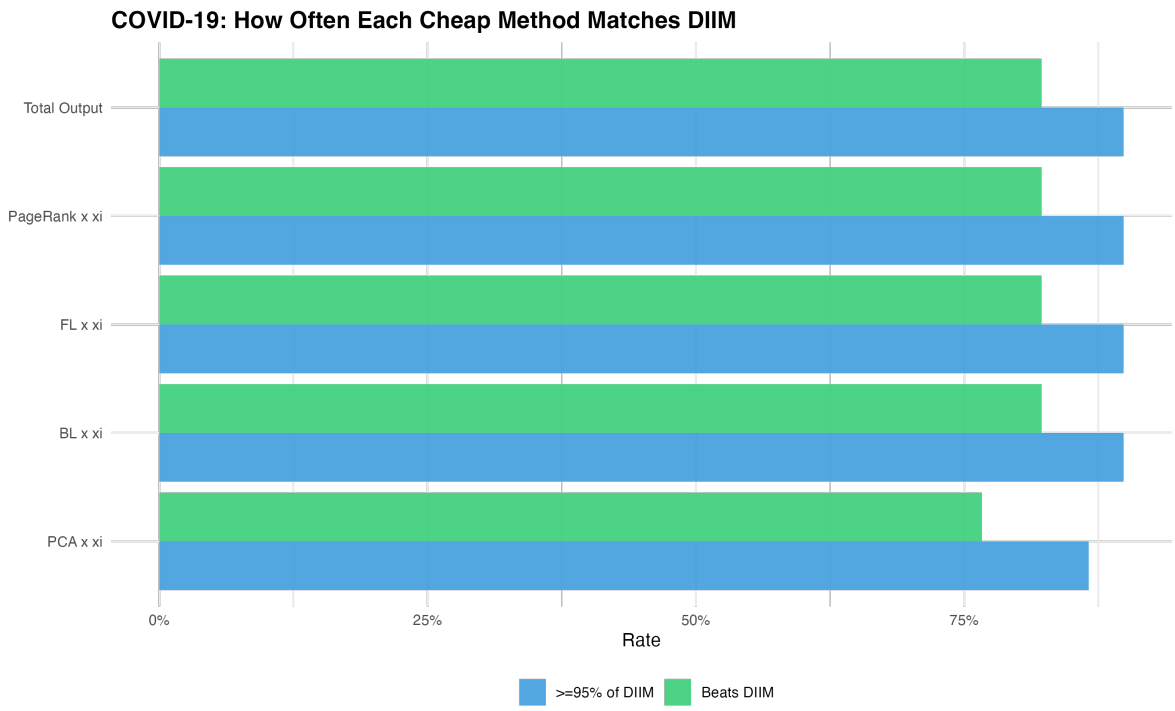

Figure 6.1: Close rates and win rates for each cheap method (COVID-19).

6.3 COVID-19 Results

Table 6.1 summarises the Monte Carlo results for the COVID-19 scenario.

Method	Close Rate ($\geq$95%)	Win Rate	Mean Ratio
Total Output	89.8%	82.2%	1.181
PageRank $\times$ xi	89.8%	82.2%	1.181
BL $\times$ xi	89.8%	82.2%	1.181
FL $\times$ xi	89.8%	82.2%	1.181
PCA $\times$ xi	86.6%	76.6%	1.119

Table 6.1: Monte Carlo results for COVID-19 (2 000 trials, $k = 5$).

Remarkably, Total Output, PageRank $\times$ xi, BL $\times$ xi, and FL $\times$ xi all achieve an identical close rate of 89.8% and a mean performance ratio exceeding 1 (i.e., they *outperform* DIIM on average). This stems from the fact that they produce the exact same top-5 sector ranking. Weighting by economic size (x_i) is the dominant factor here, allowing even the simplest baseline (Total Output) to effectively prioritise intervention in this 15-sector model. PCA $\times$ xi also performs exceptionally well with an 86.6% close rate.

Figure 6.2 shows the distribution of performance ratios across the 2 000 trials. Output-weighted linkage and PCA have distributions centred above 1, while backward linkage is centred below 0.5.

6.4 Manpower Results

Table 6.2 summarises the manpower scenario results.

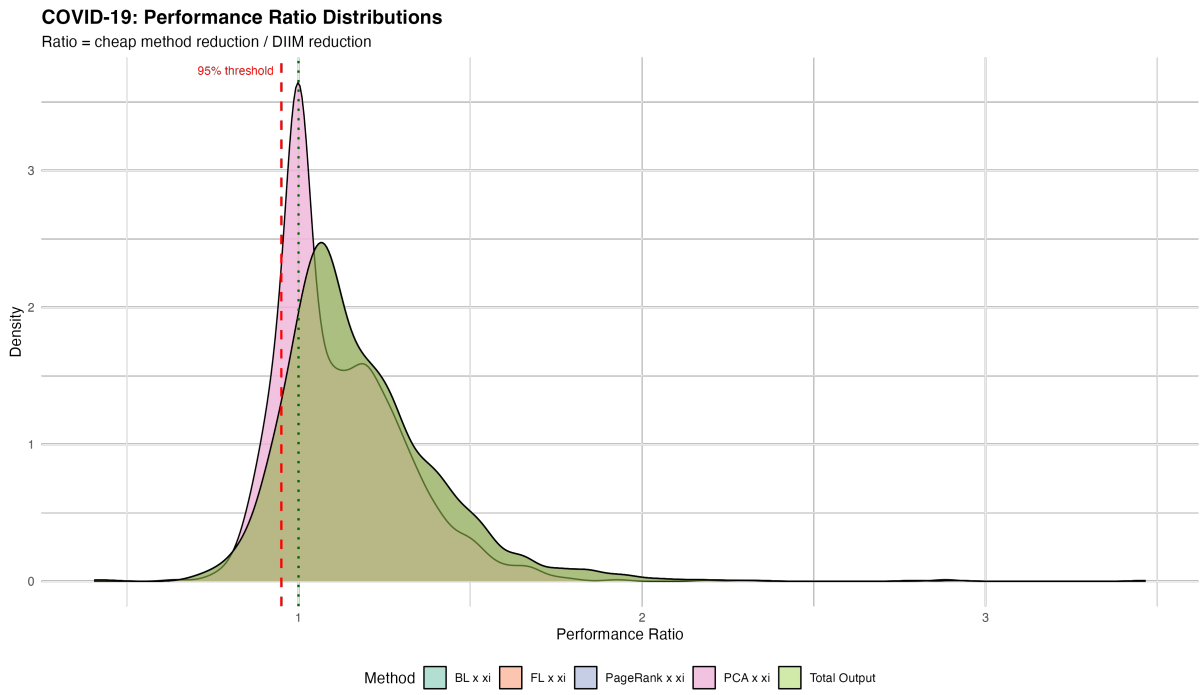

Figure 6.2: Performance ratio distributions (COVID-19).

Method	Close Rate (≥95%)	Win Rate	Mean Ratio
Total Output	60.4%	41.4%	0.953
PCA × xi	60.4%	41.4%	0.953
PageRank × xi	60.4%	41.4%	0.953
BL × xi	60.4%	41.4%	0.953
FL × xi	60.4%	41.4%	0.953

Table 6.2: Monte Carlo results for Manpower (2 000 trials, $k = 5$).

The manpower scenario is harder for all methods than the COVID-19 scenario, despite both having 15 sectors. All five cheap methods achieve an identical close rate of 60.4% and a mean performance ratio of 0.953. This identical performance stems from the near-identical top-5 sector rankings produced by all methods when weighted by economic size (x_i). The lower performance compared to COVID-19 (89.8%) is driven by the absence of the demand shock ($c^* = 0$): without c^* to anchor the disruption pattern, the DIIM's simulation-based ranking has more room to diverge from the structurally fixed rankings.

6.5 Decision Rules via Logistic Regression

For each method m, we fit a logistic regression model:

$$\log \frac{P(R_m \geq 0.95)}{1 - P(R_m \geq 0.95)} = \beta_0 + \beta_1 f_1 + \beta_2 f_2 + \cdots \tag{6.1}$$

where the features $f_1, f_2, \ldots$ are derived from q_0 and the method's sector rankings:

- **Method share**: the fraction of total q_0 captured by the method's top-5 sectors.
- **Method average rank**: the mean rank of the method's top-5 sectors in the q_0 ranking (normalised to [0,1]).

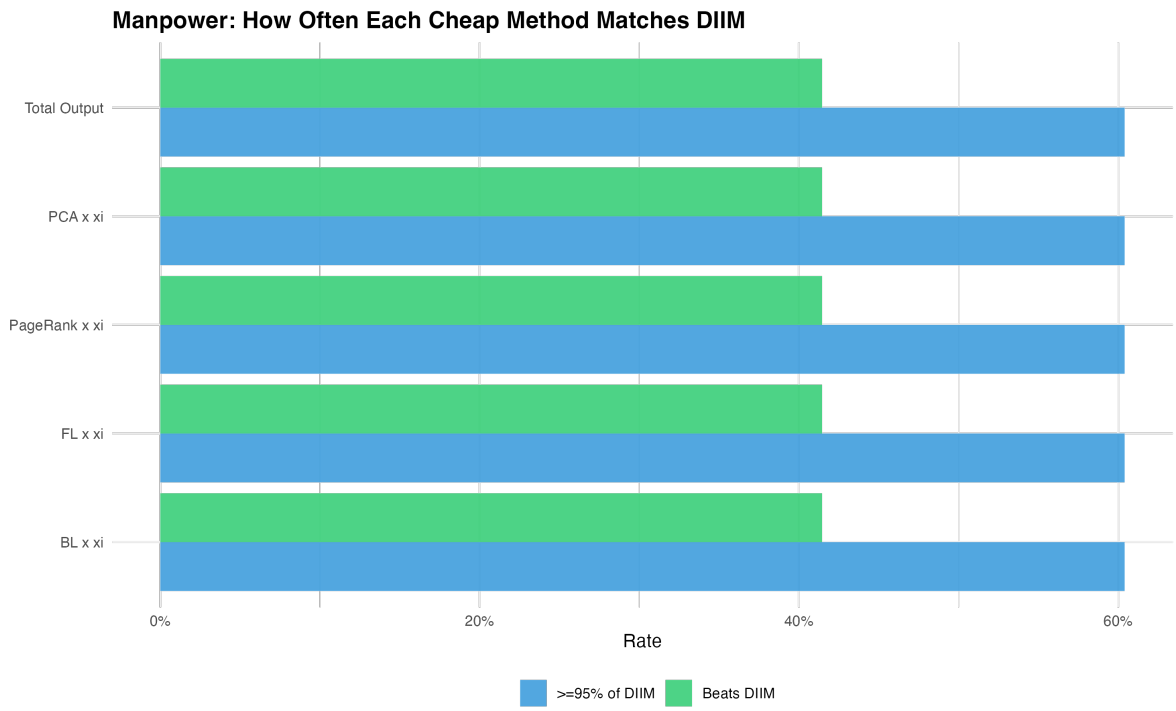

Figure 6.3: Close rates and win rates for each cheap method (Manpower).

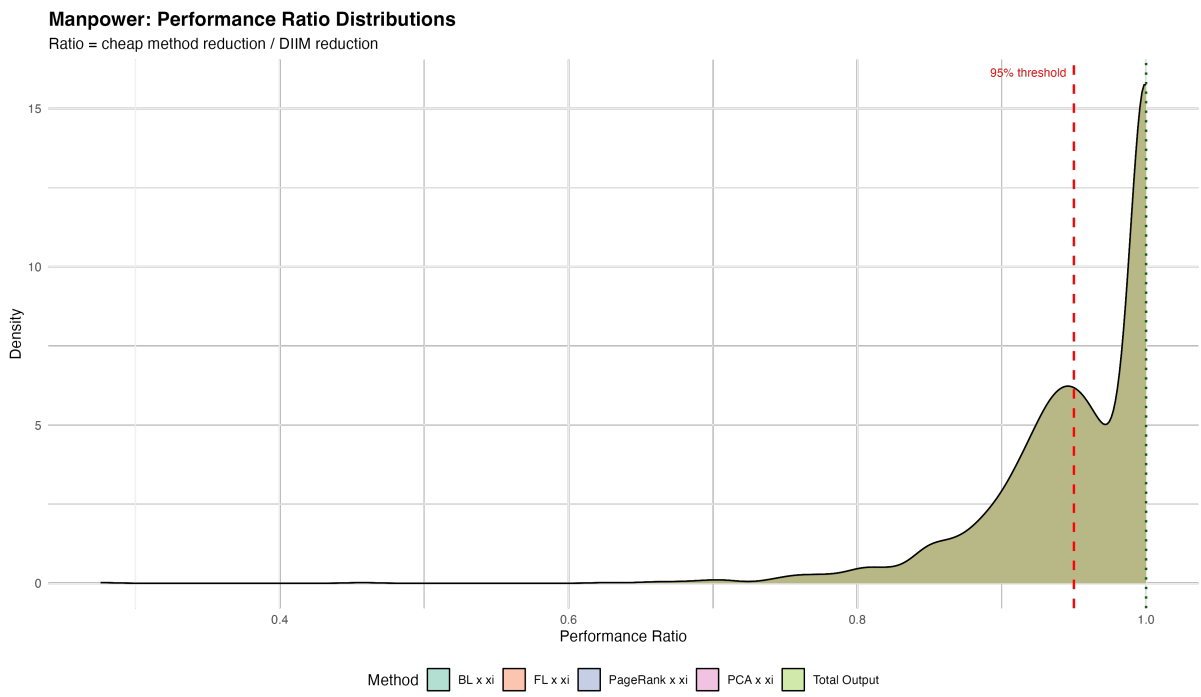

Figure 6.4: Performance ratio distributions (Manpower).

- **Overlap**: the number of the method's top-5 sectors that also appear in q_0's top-5.
- **General q_0 features**: Gini coefficient, coefficient of variation, max-to-mean ratio, and normalised entropy of q_0.

For each model, we identify the most significant predictor and compute the Youden's J optimal threshold—the threshold that maximises sensitivity plus specificity minus 1.

Table 6.3 presents the derived decision rules.

Scenario	Method	Decision Rule	J
COVID-19	Total Output	TotalOutput_avgrank < 0.533	0.402
	PageRank $\times$ xi	PageRankxi_avgrank < 0.533	0.402
	BL $\times$ xi	BLxi_avgrank < 0.533	0.402
	FL $\times$ xi	FLxi_avgrank < 0.533	0.402
	PCA $\times$ xi	PCAxi_avgrank < 0.578	0.307
Manpower	Total Output	TotalOutput_share > 0.315	0.448
	PCA $\times$ xi	PCAxi_share > 0.315	0.448
	PageRank $\times$ xi	PageRankxi_share > 0.315	0.448
	BL $\times$ xi	BLxi_share > 0.315	0.448
	FL $\times$ xi	FLxi_share > 0.315	0.448

Table 6.3: Decision rules with Youden's J-optimal thresholds.

The decision rules follow a common pattern: a cheap method works well when the disruption (q_0) is concentrated in sectors that the method has already identified as structurally important. The "share" and "average rank" features capture this alignment.

6.6 Practical Recommendations

Based on the analysis, we propose the following practical guidelines:

1. **For aggregated economies ($\leq$ 20 sectors)**: Simple baselines like Total Output or structurally-weighted methods (FL $\times$ xi, PCA $\times$ xi) can be used as primary screening tools. With close rates near 60–90%, the risk of significant underperformance is manageable, though performance varies with the shock structure (c^*).
2. **Shock structure matters**: The key driver of cheap method effectiveness at the same sectoral resolution is the shock structure. When both supply and demand shocks are present ($c^* > 0$, as in COVID-19), the demand shock anchors the disruption and cheap methods achieve close rates near 90%. Under pure supply shocks ($c^* = 0$, as in manpower disruption), close rates drop to about 60%, as the DIIM's adaptive ranking has more room to outperform fixed rankings.
3. **Pre-event planning**: Since q_0 is unknown before a disruption occurs, cheap methods are valuable for *pre-positioning* resources. The combination of structural linkage and absolute economic size provides the best generalisable estimate of which sectors are likely to be critical across a range of disruption profiles.
4. **During a crisis**: Once q_0 is observed, the decision rules in Table 6.3 can be evaluated instantly to determine whether the cheap method's recommendation is trustworthy for the specific disruption at hand.

Chapter 7

Discussion

7.1 Cross-Scenario Comparison

The two case studies reveal a clear pattern: the effectiveness of cheap key sector identification methods depends critically on the number of sectors and the structure of the IO table.

- **COVID-19 (15 sectors, $c^* > 0$)**: Total Output and all its structurally-weighted equivalents (PageRank $\times$ xi, BL $\times$ xi, FL $\times$ xi) achieve an identical close rate of nearly 90%, with PCA $\times$ xi following closely at 86.6%. With a demand shock present, the c^* vector anchors the disruption pattern, and the size-based ranking is an excellent proxy for which sectors will be most impacted under a wide range of disruption profiles.
- **Manpower (15 sectors, $c^* = 0$)**: All cheap methods achieve an identical close rate of 60.4% and mean ratio of 0.953. The close rates are notably lower than the COVID-19 scenario despite using the same number of sectors. This reveals that the shock structure—not purely the number of sectors—is the primary driver of cheap method effectiveness.

This suggests a fundamental *shock structure effect*: when both supply and demand disruptions are present ($c^* > 0$), the demand shock constrains the space of disruption outcomes, making fixed structural rankings more robust. Without c^*, the disruption dynamics are governed entirely by the purely endogenous cascading through A^*, giving the simulation-based DIIM ranking more room to diverge from any fixed ranking.

7.2 Why Economic Size Dominates Pure Structure

A defining finding of the revised analysis is that weighting structural methods by a sector's baseline economic size (x_i) overwhelmingly dominates the results. This can be understood through two mechanisms:

1. **The Anchor of Scale**: Pure structural measures (like unweighted centrality or linkage) can highlight highly interconnected sectors that are economically minuscule. Multiplying by total output ensures that structurally important but economically small sectors are not over-ranked. The massive variation in sector sizes acts as a gravity well, pulling structurally distinct methods toward nearly identical rankings.
2. **The Total Output Baseline**: Strikingly, the pure Total Output baseline (ranking sectors solely by x_i, completely ignoring the IO network) performed identically to the best structurally-weighted methods (e.g., FL $\times$ xi) in the COVID scenario, and tied for first place in the Manpower scenario. This suggests that for macro-level loss reduction, the initial scale of the intervention matters more than optimising for downstream cascading effects through complex network paths.

7.3 The Role of q_0 Concentration

The logistic regression analysis reveals that the most predictive feature across methods is the *alignment* between the method's fixed ranking and the disruption profile:

- **Methods share** (fraction of q_0 in top-k sectors): When the disruption is concentrated in sectors that the cheap method ranks highly, the method trivially works well.
- **Average rank**: When the cheap method's top-k sectors have low average rank in q_0 (i.e., they are among the most disrupted), the method performs well.

The practical implication is that the decision rules can be evaluated rapidly: once q_0 is observed, computing the share and overlap features requires only a lookup against the pre-computed structural rankings.

7.4 National Security Implications

From a national security perspective, the results support a two-stage approach to economic resilience planning:

1. **Peacetime (pre-event)**: Use Total Output or structurally-weighted methods (e.g., FL $\times$ xi) to identify sectors that are economically massive or structurally critical. These sectors should receive priority in resilience investments (e.g., stockpiling, training local replacements, diversifying supply chains) regardless of the specific disruption scenario.
2. **Crisis response**: Once the nature of the disruption is known (i.e., q_0 is observed), evaluate the decision rules. If the cheap method's conditions are met (e.g., the targeted sectors align well with the observed q_0), use its recommendations immediately. If not, run the full DIIM for a more accurate assessment. This reduces response time in the critical early hours of a crisis.

For Singapore specifically, the analysis suggests that sectors with high foreign worker concentration *and* strong IO linkages (e.g., construction, manufacturing) should be priority targets for workforce resilience measures. Even crude structural analysis using only the IO table can identify these sectors with reasonable accuracy.

7.5 Limitations

Several limitations should be noted:

1. **Uniform intervention**: All methods are compared using a standardised 10% reduction in q_0 for top-k sectors. In practice, the feasible intervention magnitude may vary across sectors.
2. **Fixed** k: The analysis primarily uses $k = 5$. Different values of k may alter the relative performance of methods.
3. **IO table vintage**: The COVID-19 scenario uses the 2019 IO table and the manpower scenario uses the 2022 IO table. Structural changes between these years are not controlled for.
4. **Recovery dynamics**: The DIIM assumes exponential recovery with fixed resilience coefficients. In practice, recovery may be non-exponential, sector-specific, and dependent on policy interventions.
5. q_0 **distribution**: The Monte Carlo analysis uses uniform random q_0 vectors. Real disruptions may have structured q_0 vectors (e.g., concentrated in specific industry groups).

7.6 Future Work

Future extensions could include:

- Incorporating structured q_0 distributions (e.g., sector-correlated disruptions) in the Monte Carlo analysis.
- Testing whether ensemble methods (combining multiple cheap method rankings) improve close rates.
- Extending the framework to multi-period disruptions where q_0 evolves over time.
- Validating the decision rules on actual disruption data from other economies.

Chapter 8

Conclusion

This project developed and applied a quantitative framework for assessing economic resilience under disruption, with direct relevance to Singapore's national security planning.

Using the Dynamic Inoperability Input-Output Model (DIIM), two disruption scenarios were studied. The COVID-19 pandemic scenario (15 sectors, 2019 IO table) demonstrated how lockdown-induced workforce unavailability cascaded through input-output linkages to produce sector-level inoperability paths and cumulative economic losses. The manpower disruption scenario (15 sectors, 2022 IO table), motivated by the risk of grey zone conflicts triggering foreign worker departures, revealed how purely structural disruption ($c^* = 0$) produces different cascading dynamics compared to a combined supply-and-demand shock.

A central contribution of this study is the systematic comparison of five cheap methods for identifying key sectors against the DIIM benchmark. Combining structural metrics with economic size (e.g., FL $\times$ xi) or simply ranking by a sector's Total Output emerged as the most robust cheap substitutes, achieving identical, high performance in 89.8% of randomised disruptions for the COVID-19 scenario and 60.4% for the manpower scenario. Notably, all five cheap methods produced identical close rates in the manpower scenario due to their near-identical sector rankings when weighted by economic size. The lower close rate under pure supply shock ($c^* = 0$) compared to the combined shock scenario ($c^* > 0$) reveals a fundamental *shock structure effect*: when a demand shock anchors the disruption pattern, fixed rankings remain robust; without it, the DIIM's adaptive ranking has more room to diverge. The remarkable equivalence in performance between pure Total Output and structurally-weighted networks underscores that a sector's absolute economic scale is the dominant predictor of macro-level loss reduction.

Decision rules were derived using logistic regression on q_0-derived features, providing policymakers with a practical tool: given observable characteristics of the disruption, determine instantly whether a cheap method's recommendation is trustworthy or whether the full DIIM should be run. For national security planning, this enables a two-stage approach—using structural analysis for peacetime resilience investments, and the decision rules for rapid crisis-time assessment.

The framework is generalisable beyond the two scenarios studied here. Any economy with an available IO table can compute the cheap structural rankings in peacetime and apply the decision rules when a disruption occurs. Future work could extend the approach to correlated disruptions, multi-period dynamics, and ensemble methods that combine multiple cheap rankings for improved robustness.

Bibliography

[1] Joost R. Santos and Yacov Y. Haimes. Modeling the demand reduction input-output (i-o) inoperability due to terrorism of interconnected infrastructures. *Risk Analysis*, 24(6):1437–1451, 2004.

[2] Jian Jin and Haoran Zhou. A demand-side inoperability input–output model for strategic risk management: Insight from the COVID-19 outbreak in shanghai, china. *Sustainability*, 15(5):4003, 2023.

[3] Ministry of Manpower, Singapore. Foreign workforce numbers. https://www.mom.gov.sg/documents-and-publications/foreign-workforce-numbers, 2023. Accessed 2026-02-17.

[4] Ministry of Health, Singapore. End of circuit breaker, phased approach to resuming activities safely. https://www.moh.gov.sg/newsroom/end-of-circuit-breaker-phased-approach-to-resuming-activities-safely/, May 2020. Newsroom release dated 19 May 2020. Accessed 2026-02-17.

[5] Ministry of Health, Singapore. Post circuit breaker measures. https://www.moh.gov.sg/newsroom/post-circuit-breaker-measures/, 2020. Newsroom / Parliamentary reply page referencing circuit breaker dates. Accessed 2026-02-17.

[6] Ministry of Defence, Singapore. Factsheet: Singapore's total defence. https://www.mindef.gov.sg/web/portal/mindef/defence-matters/defence-topic/defence-topic-detail/total-defence, 2020. Total Defence overview including economic, digital, and military defence pillars. Accessed 2026-02-17.

[7] SINGSTAT (Singapore Department of Statistics). Industry by industry input-output table (supply, use and input-output tables 2019). https://data.gov.sg/datasets/d_959da3e78690cb3136c529fbc448d178/view. data.gov.sg dataset. Accessed 2026-02-17.

Appendix A: COVID-19 Scenario Data

Sector ID	PC1	PC2
3	0.772976374	-0.158540904
2	0.488575487	0.547564001
5	0.230631604	0.602425551
8	0.173952844	0.31984955
4	0.16376151	0.197548491
10	0.141148665	0.185477914
7	0.107051197	0.232037887
11	0.100445788	0.154054165
1	0.082357584	0.203204452
9	0.065932369	0.120308858
12	0.02097817	0.033819003
6	0.01904942	0.04009044
15	0.010160651	0.015616709
14	0.004471868	0.006256627
13	0.002201633	0.00402146

Table 8.1: Sector PCA loadings (COVID-19, 15 sectors)

	Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	Manufacturing	Construction
Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	4,308.828700	2,985.482100	198.954100
Manufacturing	498.915600	59,777.284000	4,939.792300
Construction	166.973900	498.152500	23,583.027800

Table 8.2: Cross-section of technical coefficient matrix (COVID-19)

Sector ID	Sector	Initial inoperability
1	Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	0.002563
2	Manufacturing	0.000752
3	Construction	0.006890
4	Wholesale & Retail Trade	0.001639
5	Transportation & Storage	0.001107
6	Accommodation & Food Services	0.008462
7	Information & Communications	0.001936
8	Financial & Insurance Services	0.001091
9	Real Estate Services	0.000765
10	Professional Services	0.003087
11	Administrative & Support Services	0.005325
12	Public Administration & Education	0.002954
13	Health & Social Services	0.003429
14	Arts, Entertainment & Recreation	0.004552
15	Other Community, Social & Personal Services	0.015819

Table 8.3: Sector Initial Inoperability (COVID-19)

Appendix B: Manpower Scenario Data

The manpower disruption scenario uses the 2022 Singapore IO table with 107 sectors. Due to the large number of sectors, only summary statistics are presented here.

Parameter	**COVID-19**	**Manpower**
Number of sectors (n)	15	107
IO table year	2019	2022
Mean initial inoperability	0.00398	varies
Max initial inoperability	0.01582	varies
Demand shock (c^*)	> 0	$= 0$
Lockdown duration	55 days	55 days
Total horizon	751 days	751 days
PCA $\times$ xi top-5 sectors	2, 5, 4, 8, 3	57, 54, 69, 17, 26
DIIM top-5 sectors	by simulation	by simulation

Table 8.4: Summary of scenario parameters

Appendix C: Code

The R code for all simulations and analyses is available in the project repository. Key scripts include:

- `functions.R`: Core DIIM function, data loading, and cheap method implementations (Total Output, PCA $\times$ xi, PageRank $\times$ xi, BL $\times$ xi, FL $\times$ xi)
- `pca.R`: PCA analysis on the integrated IO matrix
- `simulations/decision_rules.R`: Monte Carlo framework, logistic regression, and decision boundary rules
- `simulations/enhanced_comparison.R`: Multi-PC PCA, varying top-k sweep, and baseline vs structured method comparison
- `simulations/run_simulations.R`: Parameter sweeps for severity (c^*) and lockdown duration impacts
- `manpower-disruption/manpower-disruption.R`: Manpower scenario DIIM simulation implementation